

The Vector Atlas Database User Guide

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INTRODUCTION

The Vector Atlas is an international project providing evidence-based maps to support on-the-ground malaria vector control decisions and policy. Creating these maps requires a solid evidence-base of vector spatial data to inform the underlying spatial models, necessitating the development of the Vector Atlas Database (VADB). The VADB consolidates spatial data from the published literature to create an open access analysis-ready data resource to support the project's modelling outputs and provide an open access resource for wider use.

The VADB contains *Anopheles* mosquito occurrence and bionomics (behavioural) data, as well as data on insecticide resistance, local community and environment. The current dataset includes 52 African countries, with publications from 1985 to 2023. There are currently over 41000 rows of data with confirmed sample site coordinates, with additional broadscale (polygon) data available upon request.

The VADB represents a wide array of data and as such, some consideration is needed when making use of the resource. Dedicated filter columns are provided to facilitate users' navigation to subsets of data. The database includes 'presence' and 'absence' data (i.e. where a mosquito species has been reported either as found or not found using a defined sampling method at a given location and time) [filter columns AZ or BA: 'binary_presence' OR 'binary_absence']. There are currently 37187 rows of confirmed *Anopheles* species presence and 3868 *Anopheles* species absence data rows in the VADB. There are also filters for bionomics data (data relevant to the behaviour of the vector/transmission capacity) [filter column B: 'bio_data']; insecticide resistance data (phenotypic and genotypic) [filter columns G, H or I: 'pheno_data', 'geno_data' or 'insecticide_resistance_data']; abundance data (number of specimens of a given species reported using a defined sampling method at a given location and time – NB the dataset does not report sampling effort so abundance data are not easily comparable across studies) [filter column E: 'abundance_data']. The filters 'adult_data' and 'larval_data' [filter columns C & D] indicate if adult focused and/or larval focused sampling has been reported for a specified species at a given location and time.

Key Considerations for use of the VADB

Georeferencing (spatial information)

All the data held in the VADB include the location (site name and coordinates) where the reported mosquito sample was collected. Studies report these details with various levels of precision, and this is captured in the dataset to allow users to distinguish/select the right resolution of data for their needs and are provided as categories of area type [column AJ: 'area_type']. Ranging from the finest resolution to the most coarse, area types can be 'GPS coordinates' (considered precise), 'point' (sample area <10km²); 'wide area' (sample area 10 - 25km²), 'small polygon' (25-100km²) and 'large polygon' (>100km²) (see Table 1 below). Where mosquitoes are sampled in several locations and then pooled together for analysis, the coordinates for each site are entered as a 'multiple point'. For example, if three villages were sampled and the mosquitoes pooled, the latitude and longitude for all three villages would be entered as latitude_1/longitude_1, latitude_2/longitude-2 and latitude-3/longitude_3 [columns S/T, U/V, W/X respectively]. All coordinates are given in the VADB in

decimal degrees. The VADB data download contains individual site data with single coordinates (i.e. GPS, point and wide area data). The multiple point and polygon data are available upon request.

Table 1: Area type in the site and georeferencing section of the VA dataset

Area type	Definition
GPS coords	A GPS devise has been used to record the coordinates provided.
Point	Site area (such as collections within a village) is less than 10km ² .
Wide area	Site area is between 10 to 25km ² .
Small polygon	Site area is between 25 to 100km ² . No coordinates are entered.
Large polygon	Site area is greater than 100km ² . No coordinates are entered.
Multiple points	This option is selected if multiple site locations are sampled but the mosquitoes are pooled before analysis. The coordinates for the individual sampling site are entered into columns latitude 1-8 and longitude 1-8.

Anopheles species identification

Mosquito taxonomy is complex and contains many groups of mosquitoes, referred to as species complexes. These consist of closely related sibling species that are often morphologically identical and can include dominant vector species alongside harmless non vectors.

Anopheles species identity is recorded in the VADB to the appropriate species level (species or species complex) based on the given species and whether the identification methods reported are sufficient to establish the given taxonomic level. For example, if the source refers to “*An. gambiae*” but relies on morphological identification with no clear indication whether the specimens were considered *An. gambiae* species or *An. gambiae* complex, then it is reported as *An. gambiae* complex (see the Species section for further information).

Comparability across the dataset

The VADB is a collection of data collated from a wide variety of reported studies conducted for many contrasting reasons without consistent or standardised methodologies or sampling effort. Thus, the user should always take specific care if making comparison across studies [column J: ‘source_id’]. For example, ensuring the units of a metric (e.g. the concentration used in a susceptibility bioassay or time period for the human biting rate (HBR)) are correctly filtered.

DATA RECORDS

CORE DATA

The core fields are mandatory for every source and data row and essential for full analysis and source acknowledgement. The additional fields are entered when/if those data are provided by the source.

Data Categorisation

These columns allow quick data filtering to identify rows that include data on one, some or all the categories.

Table 2: Data categorisation columns in the VADB, with descriptions

	COLUMN HEADING	DESCRIPTION
A	confidentiality_status	Denotes whether data is published, unpublished or a combination and the associated permissions for use.
B	bio_data	Data row includes bionomics data.
C	adult_data	Data row reports data collected using sampling methods targeting adult mosquitoes.
D	larval_data	Data row reports data collected using sampling methods targeting larval/pupal stage mosquitoes.
E	abundance_data	Data row includes reported number of specimens of a given species using a defined sampling method at a given location and time.
F	larval_site_data	Data row reports mosquito larval site information.
G	pheno_data	Data row includes results from an insecticide resistance bioassay (phenotypic data).
H	geno_data	Data row includes results from an insecticide resistance genetic marker test (genotypic data).
I	insecticide_resistance_data	Data row includes IR data that is phenotypic, genotypic or both. "none" indicates if no IR data is reported.

Source Information

This section contains the publication/source details to allow users to acknowledge the original source publication for the data. Every VA data source included has a unique VA assigned id number [column J: source ID] and for published sources, where possible, includes the journal Digital Object Identifier (DOI) to link back to the original publication [column K].

Table 3: Source information columns in the VADB, with descriptions

	COLUMN HEADING	DESCRIPTION
J	source_id	Unique Vector Atlas source identifier.
K	citation_doi	Unique DOI from the source. This maybe the journal article, or the DOI of a published dataset.
L	author	Coded surname of first author (ascii).
M	article_title	Coded full article title (ascii)
N	journal_title	Coded abbreviated journal title (ascii).
O	publication_year	Year the article was published.
P	personal_communication	Data row includes personal communication.

Site and Georeferencing

All VA collated data include site information with quality assessed georeferencing checked or assigned during the data abstraction process. As the prime use case for these data is spatial modelling, as much peripheral information as possible is utilised to ensure the accuracy of the

coordinates along with spatial resolution [column AJ: 'area type'] and estimated qualifications of uncertainty [column AI: 'confidence in georef']. Please also see the georeferencing on page 2 for further details.

Table 4: Site and georeferencing columns in the VADB, with descriptions

	COLUMN HEADING	DESCRIPTION
Q	country	Country reported where the study was conducted.
R	site	Site name of the location where the mosquitoes were sampled.
S	latitude_1	Latitude of primary sampling location coordinates in decimal degrees.
T	longitude_1	Longitude of primary sampling location coordinates in decimal degrees.
U	latitude_2	Latitude of second site coordinates in decimal degrees.
V	longitude_2	Longitude of second site coordinates in decimal degrees.
W	latitude_3	Latitude of third site coordinates in decimal degrees.
X	longitude_3	Longitude of third site coordinates in decimal degrees.
Y	latitude_4	Latitude of fourth site coordinates in decimal degrees.
Z	longitude_4	Longitude of fourth site coordinates in decimal degrees.
AA	latitude_5	Latitude of fifth site coordinates in decimal degrees.
AB	longitude_5	Longitude of fifth site coordinates in decimal degrees.
AC	latitude_6	Latitude of sixth site coordinates in decimal degrees.
AD	longitude_6	Longitude of sixth site coordinates in decimal degrees.
AE	latitude_7	Latitude of seventh site coordinates in decimal degrees.
AF	longitude_7	Longitude of seventh site coordinates in decimal degrees.
AG	latitude_8	Latitude of eighth site coordinates in decimal degrees.
AH	longitude_8	Longitude of eighth site coordinates in decimal degrees.
AI	confidence_in_georef	Confidence in the location of the coordinates georeferenced (within 5km, 5-10km, >10km). This field is only entered when coordinates are estimated using a map or peripheral site information in the source.
AJ	area_type	The precision of the geographical area represented by the sampling location(s)*.
AK	georef_source	The source of the site coordinate information (e.g., given in the original paper, listed in google maps, etc.).
AL	admin_level_1	The highest administration level after Country, as given by source.
AM	admin_level_2	The next administration level after Admin level_1, as given by source.

*NB: See Table 1 for more details on area type

Sampling Information

This section details the field sampling methods used to capture mosquitoes and the *Anopheles* species found. Where reported, the number of specimens found per sampling method/location/time period/species is also recorded. If a species has been reported as not found using a specified sampling method and given location/time period, a '0' is entered next to the sampling method (s) used and this is recorded as a binary absence.

Multiple sample method columns are available to capture when multiple methods of mosquito collection were used. These sampling metadata are linked to the overall occurrence of the species. Where bionomics data are also reported on the same data row (reflecting bionomics reported for the sampled population at the time/location), the sampling method(s) specific to the reported bionomic metric are recorded in the relevant bionomics section.

Table 5: Sampling information columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
AQ	sampling_occurrence_1	The primary sampling method reported in the source
AR	occurrence_n_1	The number of mosquitoes reported for sampling occurrence_1 (where given for the specified species, site and sampling period given in the data row)
AS	sampling_occurrence_2	Additional sampling method reported
AT	occurrence_n_2	The number of mosquitoes reported for sampling occurrence_2 (where given for the specified species, site and sampling period given in the data row)
AU	sampling_occurrence_3	Additional sampling method reported
AV	occurrence_n_3	The number of mosquitoes reported for sampling occurrence_3 (where given for the specified species, site and sampling period given in the data row)
AW	sampling_occurrence_4	Additional sampling method reported
AX	occurrence_n_4	The number of mosquitoes reported for sampling occurrence_4 (where given for the specified species, site and sampling period given in the data row)
AY	occurrence_n_total	The total number of mosquitoes reported for all sampling methods (where given for the specified species, site and sampling period given in the data row – this is not calculated, but represents reported information)
AZ	binary_presence	Indicates the data row refers to presence data of a species at the given time and place for any/all sampling methods used. Does not require sample numbers.
BA	binary_absence	Indicates the absence (not found) of a species at the given time and place for all sampling methods used in the study. Requires '0' for all reported methods.
BB	month_start	Month when the mosquito sampling began
BC	month_end	Month when the mosquito sampling finished
BD	year_start	Year when the mosquito sampling began
BE	year_end	Month when the mosquito sampling finished
BF	season_given	Rainy, dry or cross season at the time of the survey, as indicated in the source.
BG	season_calc	Rainy, dry or cross season at the time of the survey, as derived from information on the general seasonal timings provided from the source or calculated from other sources.
BH	rainfall_time	Level of rainfall related to the sampling period indicated in that data row (mm/month or mm/year).

Species

All data are recorded as given in the source and no assumptions are made. As these data are primarily used for species-specific spatial modelling, accurately identified *Anopheles* species are essential. Thus, to allow users to assess reliability of the reported species (and to allow us to remove obviously erroneous ones) the identification methods reported in the source are recorded.

For species from known species complexes (e.g. members of the *Anopheles gambiae* complex), an individual species will only be listed if its identification was confirmed via appropriate testing (e.g.

molecular/PCR/ chromosomal). If only morphological identification has been used, the sibling species cannot be confirmed and so the entry is listed as the species complex.

In 2013, *An. gambiae* was officially split into two formally named species, previously given as Form M - now *An. coluzzii*; and Form S - now *An. gambiae*. As the VA dataset contains data from before 2013, to ensure we accurately represent the data and species as given in the source, the species list includes 'gambiae_s form_m form', indicating where the reported species has not been clearly classified to either *An. coluzzii* or *An. gambiae*.

Table 6: *Anopheles* species columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
BI	species	The <i>Anopheles</i> species, species complex or subgroup.
BJ	species_id_1	The primary method used to identify the <i>Anopheles</i> species.
BK	species_id_2	The secondary or more in-depth method used to identify the <i>Anopheles</i> species.

ADDITIONAL FIELDS – COMMUNITY AND ENVIRONMENT

The following sections are only completed if the source report the information in direct relation to the specific sampling area during the sampling period. I.e. general district wide descriptions or background information is not entered.

Housing

This section records data reported in the source that relate to the majority of houses in the study area.

Table 7: Housing columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
BL	roof	Reported roof material (metal, thatch, mix, other) of all (or the majority) of houses in the study area. 'Mix' if there is an equal mix of houses with thatch and metal roofs in the study area. "Other" indicates all other roof types.
BM	walls	Reported wall construction material (brick, mud, mix, other) of all (or the majority) of houses in the sampling area. 'Mix' if there is an equal mix of houses with brick or mud walls in the study area. Other" indicates all other wall types.
BN	house_screening	Source indicates the majority of homes in the study area have/do not have screening (on windows and doors).
BO	open_eaves	Source indicates the majority of homes in the study area have/do not have open eaves.
BP	cooking	Source indicates that the householders in the survey area mostly cook indoors, outdoors or peridomestically.

Community

This section includes information relevant to community behaviour that could increase or decrease human exposure to mosquito bites.

Table 8: Community columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
BQ	common_occupation_1	Primary/most common occupation of the community in the study area reported in the source
BR	common_occupation_2	Second most common occupation of the community in the study area reported in the source
BS	common_occupation_3	Third most common occupation of the community in the study area reported in the source
BT	outdoor_activities_at_night	Yes/no/na. Outdoor community activities outdoors (e.g. socialising) when mosquitoes are active
BU	sleeping_outdoors	Indicates where the source reports that community members regularly do/do not sleep outdoors in the given sampling period/site
BV	outdoor_timings_hours	The average number of hours community members spend outdoors in the given sampling period/site
BW	average_bedtime	The average bedtime of the community in the reported site/survey time period (24-hour clock format)
BX	average_wake_time	The average awake time of the community in the reported site/survey time period (24-hour clock format)
BY	time_people_leave_home_in_morning	The average time members of the community in the reported site/survey time period leave their homes (24-hour clock format)
BZ	hours_spent_away_from_home_per_day	The average number of hours members of the community in the reported site/survey time period are away from home during the day
CA	seasonal_labour	Yes/no/na relating to the community at the reported site/survey time period

Environment

This section records reported environmental and ecological information reported in the source.

Table 9: Environment columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
CB	forest	The presence of forest in or adjacent to the sampling location.
CC	farming	The presence of farmed crops (e.g. rice) in the sampling location
CD	livestock_1	Primary species of livestock reported at the reported present in the sampling site/survey period.
CE	livestock_2	Second most common species of livestock reported at the reported present in the sampling site/survey period.
CF	livestock_3	Third most common species of livestock reported at the reported present in the sampling site/survey period.
CG	livestock_4	Fourth most common species of livestock reported at the reported present in the sampling site/survey period.
CH	local_plants	List of all the species of plants reported as present at the site and sampling period or the study.

ADDITIONAL FIELDS – ANOPHELES BIONOMICS

The Bionomics data reports measured metrics specific to the behaviour (e.g. biting location) or biology (e.g. parity) of the mosquitoes sampled at the given time and location that are of specific relevance to the capacity of the vector to transmit malaria. Each section links the mosquito sampling method to the bionomic metric as often the sampling method is key in either measuring a certain mosquito characteristic (e.g., indoor or outdoor biting) or allows a level of quality control, i.e., if the sampling method is not suitable for generating data for the metric being reported.

Biology

This section contains data regarding mosquito characteristics, life cycle and role as disease vectors.

Table 10: *Anopheles* biology columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
CI	sampling_biology_1	Primary sampling method used to collect the mosquitoes used to determine the reported biological
CJ	sampling_biology_2	Second sampling method used to collect the specimens detailed in the Biology section.
CK	sampling_biology_3	Third sampling method used to collect the specimens detailed in the Biology section.
CL	sampling_biology_n	"Yes" indicates that there are more than three sampling methods.
CM	parity_n	The number of parous females detected from the total number examined.
CN	parity_total	The total number of females examined for parity.
CO	parity_percent	The percentage of parous females in the sample: = number of parous females/total number examined x 100.
CP	daily_survival_rate_percent	The estimated proportion of female mosquitoes alive on day d that are still alive on day d+1.
CQ	fecundity_mean_batch_size	The number of eggs laid per batch.
CR	gonotrophic_cycle_days	The number of days for a female mosquito to go through the reproduce-feeding cycle.

Infection

This section records metrics of *Plasmodium* infection within the vector.

Table 11: Infection columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
CS	sampling_infection_1	Primary sampling method used to collect the specimens detailed in the Infection section.
CT	sampling_infection_2	The second sampling method used to collect the specimens detailed in the Infection section.
CU	sampling_infection_3	The third sampling method used to collect the specimens detailed in the Infection section.
CV	sampling_infection_n	"Yes" indicates that more than three sampling methods were reported in the source.
CW	sporozoite_rate_by_dissection_n	The number of sporozoite positive mosquitoes detected by dissection.

CX	sporozoite_rate_by_dissection_total	The total number of mosquitoes dissected.
CY	sporozoite_rate_by_dissection_percent	The sporozoite rate (SR) (percent) = $SR_n / SR_{total} \times 100$ from dissection.
CZ	sporozoite_rate_by_csp_n_pool	The number of sporozoite positive mosquitoes detected by circumsporozoite protein (CSP) in a sample (pool).
DA	sporozoite_rate_by_csp_total_pool	The total number of mosquitoes in the sample (pool) tested by CSP.
DB	sporozoite_rate_by_csp_percent	The sporozoite rate (percent) = $SR_{n_pool} / SR_{total_pool} \times 100$ from CSP.
DC	sporozoite_rate_p_falciparum_n	The number of <i>P. falciparum</i> sporozoite positive mosquitoes detected by CSP in a sample.
DD	sporozoite_rate_p_falciparum_total	The total number of mosquitoes in the sample (pool) tested by CSP for <i>P. falciparum</i> sporozoites.
DE	sporozoite_rate_p_falciparum_percent	The <i>P. falciparum</i> sporozoite rate (percent) = $PfSR_n / PfSR_{total} \times 100$ from CSP.
DF	sporozoite_rate_p_vivax_n	The number of <i>P. vivax</i> sporozoite positive mosquitoes detected by CSP in a sample.
DG	sporozoite_rate_p_vivax_total	The total number of mosquitoes in the sample (pool) tested by CSP for <i>P. vivax</i> sporozoites.
DH	sporozoite_rate_p_vivax_percent	The <i>P. vivax</i> sporozoite rate (percent) = $PvSR_n / PvSR_{total} \times 100$ from CSP.
DI	oocyst_n	The number of oocyst infected females detected from the total number examined.
DJ	oocyst_total	The total number of females examined for oocysts.
DK	oocyst_rate_percent	The percentage of oocyst infected females detected in the sample: = number of infected females/total number examined x 100.
DL	eir	The entomological inoculation rate (EIR) - the number of infectious mosquito bites an individual receives over a specific period (e.g., per person per night, month, or year) – the product of HBR and SR.
DM	eir_period	The unit of time relating to the EIR (day, week, month, year).
DN	ext_incubation_period_days	The extrinsic (ext) incubation period of the malaria parasite in days.

Biting rate

This section records the biting rate – the frequency of mosquito bites to either humans or animals per unit time.

Table 12: Biting rate columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
DO	hbr_sampling_indoor	The sampling method used to collect the mosquitoes from which indoor human biting rate is evaluated.
DP	indoor_hbr	The indoor human biting rate; the number of bites received indoors per person per unit time.
DQ	hbr_sampling_outdoor	The sampling method used to collect the mosquitoes from which outdoor human biting rate is evaluated.
DR	outdoor_hbr	The outdoor human biting rate; the number of bites received outdoors per person per unit time.

DS	hbr_sampling_combined_1	The first sampling method used to collect the mosquitoes from which human biting rate (HBR) is evaluated where data are amalgamated from more than one method.
DT	hbr_sampling_combined_2	The second sampling method used to collect the mosquitoes from which human biting rate is evaluated where data are amalgamated from more than one method.
DU	hbr_sampling_combined_3	The third sampling method used to collect the mosquitoes from which human biting rate is evaluated where data are amalgamated from more than one method.
DV	hbr_sampling_combined_n	"Yes" indicates that more than three sampling methods were reported in the source.
DW	combined_hbr	The human biting rate evaluated from the data from amalgamated sampling methods.
DX	hbr_unit	The unit time for the human biting rate data.
DY	abr_sampling_1	The first sampling methods used to collect the mosquitoes from which the animal biting rate (ABR) is evaluated where data are amalgamated from more than one method.
DZ	abr_sampling_2	The second sampling methods used to collect the mosquitoes from which the animal biting rate is evaluated where data are amalgamated from more than one method.
EA	abr_sampling_3	The third sampling methods used to collect the mosquitoes from which the animal biting rate is evaluated where data are amalgamated from more than one method.
EB	abr_sampling_n	"Yes" indicates that more than three sampling methods were reported in the source
EC	abr	The animal biting rate evaluated from the data from amalgamated sampling methods.
ED	abr_unit	The unit time for the animal biting rate data.

Host preferences

This section records the host (human or animal) reported by the source.

Table 13: Host preference columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
EE	host_sampling_indoor	The indoor sampling method used to collect the mosquitoes from which indoor host preference is evaluated.
EF	indoor_host_n	The number of mosquitoes positively indicating a measure of host preference from the total number collected indoors.
EG	indoor_host_total	The total number of mosquitoes sampled indoors examined for measures of host preference
EH	indoor_host_percent	The percent measure of host preference from indoor sampled mosquitoes.
EI	host_sampling_outdoor	The outdoor sampling method used to collect the mosquitoes from which outdoor host preference is evaluated
EJ	outdoor_host_n	The number of mosquitoes positively indicating a measure of host preference from the total number collected outdoors.
EK	outdoor_host_total	The total number of mosquitoes sampled outdoors examined for measures of host preference
EL	outdoor_host_percent	The percent measure of host preference from outdoor sampled

		mosquitoes
EM	host_sampling_combined_1	The first sampling method used to collect the mosquitoes from which host preference is evaluated (amalgamated from more than one sampling method).
EN	host_sampling_combined_2	The second sampling method used to collect the mosquitoes from which host preference is evaluated (amalgamated from more than one sampling method).
EO	host_sampling_combined_3	The third sampling method used to collect the mosquitoes from which host preference is evaluated (amalgamated from more than one sampling method).
EP	host_sampling_combined_n	"Yes" indicates that more than three sampling methods were reported in the source
EQ	combined_host_n	The number of mosquitoes positively indicating a measure of host preference collected by a combination of sampling methods.
ER	combined_host_total	The total number of mosquitoes amalgamated from more than one sampling method, examined for measures of host preference.
ES	combined_host	The measure of host preference from mosquitoes amalgamated from more than one sampling method.
ET	host_unit	Indicates the measure/unit used to identify host preference.
EU	host_sampling_other_1	Where additional data (other than captured in previous host columns) are presented examining host preference, they are recorded here ('other' columns). The first sampling methods used to collect the mosquitoes from which host preference is evaluated (additional data).
EV	host_sampling_other_2	The second sampling methods used to collect the mosquitoes from which host preference is evaluated (additional data).
EW	host_sampling_other_3	The third sampling methods used to collect the mosquitoes from which host preference is evaluated (additional data).
EX	host_sampling_other_n	"Yes" indicates that more than three sampling methods were reported in the source
EY	other_host_n	The number of mosquitoes positively indicating a measure of host preference (additional data).
EZ	other_host_total	The total number of mosquitoes examined for measures of host preference (additional data).
FA	host_other	The measure of host preference (additional data).
FB	host_other_unit	Indicates the measure used to identify host preference (additional data).

Biting location

This section indicates the vector's preference for indoor or outdoor biting.

Table 14: Biting location columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
FC	biting_number_of_sampling_nights_indoors	The sampling effort, in 'number of person nights', to collect the indoor biting data.
FD	biting_sampling_indoor	The sampling method used to collect the indoor mosquitoes from which biting location preference is determined.
FE	indoor_biting_n	The number of mosquitoes found biting indoors.
FF	indoor_biting_total	The total number of indoor and outdoor biting mosquitoes.

FG	indoor_biting_data	The percentage or ratio of mosquitoes found biting indoors.
FH	biting_number_of_sampling_nights_outdoors	The sampling effort, in 'number of person nights', to collect the outdoor biting data.
FI	biting_sampling_outdoor	The sampling method used to collect the outdoor mosquitoes from which biting location preference is determined.
FJ	outdoor_biting_n	The number of mosquitoes found biting outdoors.
FK	outdoor_biting_total	The total number of indoor and outdoor biting mosquitoes.
FL	outdoor_biting_data	The percentage or ratio of mosquitoes found biting outdoors.
FM	indoor_outdoor_biting_unit	The unit of the indoor and/or outdoor biting data (e.g. %, ratio).

Biting activity peak

This section records vector 'activity' peaks (hourly biting density) in- or outdoors or where the data is reported for in and out combined (combined).

Table 15: Biting activity peak columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
FN	biting_activity_indoor_number_of_sampling_nights	The sampling effort, in 'number of person nights', relevant to indoor biting activity data.
FO – FZ	[individual columns for each hour between 18:00 and 06:00 for indoor biting peak]	"1" is given during the hour where mosquito density peaked indoors.
GA	biting_activity_outdoor_number_of_sampling_nights	The sampling effort, in 'number of person nights', relevant to outdoor biting activity data.
GB – GM	[individual columns for each hour between 18:00 and 06:00 for outdoor biting peak]	"1" is given during the hour where mosquito density peaked outdoors.
GN	biting_activity_combined_number_of_sampling_nights	The sampling effort, in 'number of person nights', relevant to combined indoor and outdoor biting activity data.
GO – GZ	[individual columns for each hour between 18:00 and 06:00 for combined biting peak]	Where in and outdoor mosquito density data are reported combined, "1" is given during the hour where mosquito density peaked.

Resting preference

This section records data indicating where mosquitoes rest after blood feeding and their fed/ gravid status.

Table 16: Resting preference columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
HA	resting_sampling_indoor	Indoor sampling method used to collect the mosquitoes to assess indoor resting behaviour.
HB	unfed_indoor	Total number of unfed mosquitoes reported in the sample collected indoors.
HC	fed_indoor	Total number of fed mosquitoes reported in the sample collected indoors.
HD	gravid_indoor	Total number of gravid mosquitoes reported in the sample collected indoors.
HE	total_indoor	Total number of mosquitoes reported in the sample collected indoors, including unfed, fed, and gravid females.

HF	resting_sampling_outdoor	Outdoor sampling method used to collect the mosquitoes to assess outdoor resting behaviour.
HG	unfed_outdoor	Total number of unfed mosquitoes reported in the sample collected outdoors.
HH	fed_outdoor	Total number of fed mosquitoes reported in the sample collected outdoors.
HI	gravid_outdoor	Total number of gravid mosquitoes reported in the sample collected outdoors.
HJ	total_outdoor	Total number of mosquitoes reported in the sample collected outdoors, including unfed, fed, and gravid females.
HK	resting_sampling_other	Sampling methods relating to 'other' data. Columns appended '_other' are used when additional sampling is reported, for example if in- and outdoor resting mosquitoes are listed in the previous sections, but the source also reports data from a third sampling method (e.g. animal sheds, or peri-domestic).
HL	unfed_other	Total number of unfed mosquitoes reported in the sample collected by additional/'other' methods.
HM	fed_other	Total number of fed mosquitoes reported in the sample collected by additional/'other' methods.
HN	gravid_other	Total number of gravid mosquitoes reported in the sample collected by additional/'other' methods.
HO	total_other	Total number of mosquitoes in the sample collected by additional/'other' methods, including unfed, fed, and gravid females.
HP	resting_unit	The data unit relating to the indoor, outdoor, or other resting data on the data row.

Larval site

This section captures larval site characteristics for up to three reported larval sites.

Table 17: Larval site columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
HQ	larval_instars_found_1*	The different larval instar(s) and pupae reported in the first sampled larval habitat.
HR	larval_habitat_1	Description of the larval site.
HS	larval_site_character_1	Larval site categorisation (natural or artificial).
HT	larval_turbidity_1	Larval site categorisation (turbid, clear, polluted).
HU	larval_salinity_1	Larval site categorisation (brackish or fresh).
HV	larval_vegetation_1	Larval site categorisation (vegetation present, absent, or partial coverage).
HW	larval_shade_1	Larval site categorisation (shaded, sunlit or partially shaded).
HX	larval_water_current_1	Larval site categorisation (water flow is slow, fast or stagnant).
HY	larval_size_1	Diameter of the larval site at its widest point (cm/m).
HZ	larval_depth_1	Depth of the larval site (cm).
IA	larval_permanence_1	Larval site categorisation (permanent, temporary or seasonal).
IB	larval_other_fauna_1	Larval site categorisation (presence of other fauna categorised as predator or competitor).
IC	larval_control_present_1	If larval control is present and whether it is physical or chemical control.

*These columns are repeated in the template with “_2” [columns ID – IP] and “_3” [columns IQ – JC] for additional habitats represented in a single pooled sample of mosquitoes.

ADDITIONAL FIELDS – INSECTICIDE RESISTANCE (IR)

IR Based Control

This section indicates if any insecticide based control measures (e.g. LLIN distribution) have been deployed recently (as reported in the source) at each reported sampling site/data row.

Table 18: IR Based Control columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
AN	insecticide_control	Indicates whether the source reports the presence of insecticide-based control (previously implemented or implemented as part of the referenced study) at the specified location and time period given in the data row.
AO	control_type	If the source reports the presence of insecticide control, the method of insecticide control is recorded.
AP	itn_use	Records actual reported use of ITN (i.e. beyond ownership) when provided in the source.

Insecticide resistance bioassays

This section records data on insecticide resistance via susceptibility tests, including standard (discriminating dose), intensity bioassays and synergist tests.

Table 19: IR Bioassay columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
JD	bioassay_representative_of_complex_at_site	‘Yes’ if the mosquito sample used in the susceptibility test is representative of the species complex at the sampling site/period.
JE	bioassay_represents_complex_if_disaggregated_values_combined	‘Yes’ if multiple rows of data for individual sibling species at the given site/time period can be combined to give a representative result for the species complex.
JF	generation	The mosquito generation tested: F0, F1 or a mix of both.
JG	wild_caught_larvae_or_adults	Reports the mosquito life stage sampled.
JH	lower_age_days	Lower age of mosquito age range within a sample used in susceptibility tests (days).
JI	upper_age_days	Upper age of mosquito age range within a sample used in susceptibility tests (days).
JJ	test_protocol	The WHO or CDC bioassay protocol followed and year.
JK	insecticide_tested	The insecticide used in the susceptibility test on the mosquito sample from the given site/time period in the data row.
JL	insecticide_class	Chemical class of the insecticide under evaluation.
JM	irac_moa	Insecticide Resistance Action Committee (IRAC) mode of action (MoA) recorded for the insecticide under evaluation.
JN	irac_moa_code	The IRAC MoA code for the insecticide under evaluation.
JO	concentration_percent	Insecticide concentration tested, given as a percent (WHO).

JP	concentration_micrograms	Insecticide concentration tested, given in µg/bottle (CDC).
JQ	exposure_period_min	The reported period of time the mosquitoes were exposed to the insecticide in the susceptibility test (minutes).
JR	intensity_multiplier	The number by which the recommended or baseline insecticide concentration has been multiplied where greater dose concentrations have been tested. "1" = baseline/standard concentration.
JS	synergist_tested	When tested, the synergist tested alongside the insecticide.
JT	synergist_concentration	The given concentration of the synergist.
JU	synergist_concentration_unit	The synergist concentration unit.
JV	mosquitoes_tested_n	The total number of mosquitoes tested across all replicates.
JW	mosquitoes_dead_n	The total number of mosquitoes reported dead across all replicates.
JX	percent_mortality	The percentage of dead mosquitoes reported across all replicates.
JY	knock_down_exposure_time_min	The period of exposure to the insecticide in minutes before knock down recorded.
JZ	mosquitoes_knocked_down_n	Number of mosquitoes knocked down at recorded exposure time.
KA	knock_down_percent	Percentage of mosquitoes reported knocked down at the given exposure time.
KB	kdt_50_percent_min	The time taken to knock down 50% of the mosquito sample (minutes).
KC	kdt_90_percent_min	The time taken to knock down 90% of the mosquito sample (minutes).
KD	kdt_95_percent_min	The time taken to knock down 95% of the mosquito sample (minutes).

Genotypic representativeness

This section records how representative the mosquito tested for genetic mechanisms of resistance are to the sampled population and/or those used in the insecticide resistance bioassays.

Table 20: Genotypic representativeness columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
KE	genotypic_test_representative_of_species_at_site	"Yes" indicates that the mosquitoes used to identify the genotypic IR mechanism are representative of the sampled species population at the reported site/sample period.
KF	genotypic_represents_species_if_disaggregated_values_combined	Multiple rows of data would be representative if disaggregated values were combined without adjustments.
KG	minor_species_missing_allele_frequency_data	"Yes" indicates if minor species identified but genetic mechanism data for that species is not reported.
KH	genotypic_sample_first_been_through_bioassay_tests	"Yes" if sample first exposed to bioassay testing before being selected for genotypic tests. Including pooled samples from multiple bioassays.
KI	genotypic_sample_linked_to_a_specific_bioassay	"Yes" if the sample used in the genetic mechanisms tests matches to a specific bioassay test (not a pooled sample but linked to a specific insecticide tested).
KJ	bioassay_subsample_used	Indicates the subsample of mosquitoes from the bioassay test

_in_genotypic_test	were utilised in genotypic tests: dead only, alive only, dead and alive or not specified.
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Insecticide resistance genetic markers

The VADB collates data for five genetic markers including target-site resistance for Voltage-gated sodium channels (Vgsc) (formally known as *kdr*), resistance to dieldrin locus (*Rdl*) (GABA receptor), altered Acetylcholinesterase (Ace-1) plus metabolic resistance for Glutathione S-transferases (Gste) and Cytochrome P450s (Cyp).

Voltage gated sodium channel (Vgsc) markers

This section records the voltage gated sodium channel (Vgsc) markers: Vgsc-995S (formally L1014S, *kdr*-East) and Vgsc-995F (formally L1014F, *kdr*-West), Vgsc-402 and Vgsc-1570 loci. All publications using previous terms are recorded using the current nomenclature.

Table 21: Voltage gated sodium channel (Vgsc) columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
KK	vgsc_method_1	First method used to test for the vgsc mutation(s) in the format Author Year of publication.
KL	vgsc_method_2	Second method used to test for the vgsc mutation(s) in the format Author Year of publication.
KM	vgsc_number_of_mosquitoes_tested	The total number of mosquitoes in the sample used for all the vgsc tests.
KN	vgsc_generation	The mosquito generation tested: F0 (wild caught), F1 (first generation lab reared) or a mix of both.
<i>Vgsc995 genotype frequencies</i>		
KO	vgsc995l_vgsc995l_n	The number of mosquitoes in the sample with the 995L/995L genotype (homozygous susceptible (SS)).
KP	vgsc995l_vgsc995l_percent	Frequency (percent) of mosquitoes in the sample with the 995L/995L genotype.
KQ	vgsc995l_vgsc995f_n	The number of mosquitoes in the sample with the 995L/995F genotype (heterozygous resistant (RS)).
KR	vgsc995l_vgsc995f_percent	Frequency (percent) of mosquitoes in the sample the 995L/995F genotype.
KS	vgsc995f_vgsc995f_n	The number of mosquitoes in the sample with the 995F/995F genotype (homozygous resistant (RR)).
KT	vgsc995f_vgsc995f_percent	Frequency (percent) of mosquitoes in the sample with the 995F/995F genotype.
KU	vgsc995l_vgsc995s_n	The number of mosquitoes in the sample with the 995L/995S genotype (heterozygous resistant (RS)).
KV	vgsc995l_vgsc995s_percent	Frequency (percent) of mosquitoes in the sample with the 995L/995S genotype.
KW	vgsc995s_vgsc995s_n	The number of mosquitoes of mosquitoes in the sample with the 995S/995S genotype (homozygous resistant (RR)).
KX	vgsc995s_vgsc995s_percent	Frequency (percent) of mosquitoes in the sample with the 995S/995S genotype.
KY	vgsc995l_vgsc995c_n	The number of mosquitoes in the sample with the 995L/995C genotype (heterozygous resistant (RS)).
KZ	vgsc995l_vgsc995c_percent	Frequency (percent) of mosquitoes in the sample with the 995L/995C genotype.

LA	vgsc995c_vgsc995c_n	The number of mosquitoes in the sample with the 995C/995C genotype (homozygous resistant (RR)).
LB	vgsc995c_vgsc995c_percent	Frequency (percent) of mosquitoes in the sample with the 995C/995C genotype.
LC	null_vgsc995c_or_vgsc995c_vgsc995c_n	The number of mosquitoes in the sample with the null/995C genotype (null = non-functional allele).
LD	null_vgsc995c_or_vgsc995c_vgsc995c_percent	Frequency (percent) of mosquitoes in the sample with the null/995C genotype.
LE	vgsc995f_vgsc995s_n	The number of mosquitoes in the sample with the 995F/995S genotype (homozygous resistant (RR)).
LF	vgsc995f_vgsc995s_percent	Frequency (percent) of mosquitoes in the sample with the 995F/995S genotype.
LG	vgsc995f_vgsc995c_n	The number of mosquitoes in the sample with the 995F/995C genotype (homozygous resistant (RR)).
LH	vgsc995f_vgsc995c_percent	Frequency (percent) of mosquitoes in the sample with the 995F/995C genotype.
<i>SS/SR/RR provided without specifying mutation</i>		
LI	susceptible_susceptible_n	The number of mosquitoes in the sample when vgsc/kdr data is provided in general terms of susceptible/susceptible genotype without the specific mutation being reported (homozygous susceptible (SS)).
LJ	susceptible_susceptible_percent	Frequency (percent) of mosquitoes in the sample with the susceptible/susceptible genotype.
LK	resistant_susceptible_n	The number of mosquitoes in the sample when vgsc/kdr data is provided in general terms of a resistant/susceptible genotype without the specific mutation being reported (heterozygous resistant (RS)).
LL	resistant_susceptible_percent	Frequency (percent) of mosquitoes in the sample with the resistant/susceptible genotype.
LM	resistant_resistant_n	The number of mosquitoes in the sample when vgsc/kdr data is provided in general terms of a resistant/resistant genotype without the specific mutation being reported (homozygous resistant (RR)).
LN	resistant_resistant_percent	Frequency (percent) of mosquitoes in the sample with the resistant/resistant genotype.
<i>Vgsc995 allele frequencies</i>		
LO	vgsc995l_percent	Frequency (percent) of mosquitoes in the sample with the 995L (susceptible) allele.
LP	vgsc995f_percent	Frequency (percent) of mosquitoes in the sample with the 995F (resistant) allele.
LQ	vgsc995s_percent	Frequency (percent) of mosquitoes in the sample with the 995S (resistant) allele.
LR	vgsc995c_percent	Frequency (percent) of mosquitoes in the sample with the 995C (resistant) allele.
LS	kdr_percent	Frequency (percent) mosquitoes in the sample with if allele frequency given as 'kdr' without the specific mutation.
<i>Vgsc402 genotype frequencies</i>		
LT	vgsc402v_vgsc402v_n	Number of mosquitoes in the sample with the 402/402 genotype (homozygous susceptible (SS)).
LU	vgsc402v_vgsc402v_percent	Frequency (percent) of mosquitoes in the sample with the 402/402 genotype.

LV	vgsc402v_vgsc402l_n	Number of mosquitoes in the sample with the 402/402L genotype (heterozygous resistant (RS)).
LW	vgsc402v_vgsc402l_percent	Frequency (percent) of mosquitoes in the sample with the 402/402L genotype.
LX	vgsc402l_vgsc402l_n	Number of mosquitoes in the sample with the 402L/402L genotype (homozygous resistant (RR)).
LY	vgsc402l_vgsc402l_percent	Frequency (percent) of mosquitoes in the sample with the 402L/402L genotype.
Vgsc402 allele frequencies		
LZ	vgsc402v_percent	Frequency (percent) of mosquitoes in the sample with the Vgsc-402 (susceptible) allele.
MA	vgsc_402l_percent	Frequency (percent) of mosquitoes in the sample with the Vgsc-402L (resistant).
Vgsc1570 genotype frequencies		
MB	vgsc1570n_vgsc1570n_n	Number of mosquitoes in the sample with the 1570/1570 genotype (homozygous susceptible (SS)).
MC	vgsc1570n_vgsc1570n_percent	Frequency (percent) mosquitoes in the sample with the 1570/1570 genotype.
MD	vgsc1570n_vgsc1570y_n	Number of mosquitoes in the sample with the 1570/1570Y genotype (heterozygous resistant (RS)).
ME	vgsc1570n_1570y_percent	Frequency (percent) of mosquitoes in the sample with the 1570/1570Y genotype.
MF	vgsc1570y_vgsc1570y_n	Number of mosquitoes in the sample with the 1570Y/1570Y genotype (homozygous resistant (RR)).
MG	vgsc1570y_vgsc1570y_percent	Frequency (percent) of mosquitoes in the sample with the 1570/1570Y genotype.
Vgsc1570 allele frequencies		
MH	vgsc1570n_percent	Frequency (percent) of mosquitoes in the sample with the 1570 (susceptible) allele.
MI	vgsc1570y_percent	Frequency (percent) of mosquitoes in the sample with the 1570Y (resistant) allele.

Resistance to dieldrin (Rdl) markers

This section records two mutations associated with resistance to dieldrin locus (*Rdl*): rdl-296G and rdl-296S.

Table 22: Resistance to dieldrin (*Rdl*) columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
MJ	rdl_method_1	Method used for the Resistance to dieldrin locus (<i>Rdl</i>) test in the format Author Year of publication.
MK	rdl_number_of_mosquitoes_tested	The total number of mosquitoes in the sample used for the <i>Rdl</i> test.
ML	rdl_generation	The mosquito generation tested: F0, F1 or a mix of both.
Rdl296 genotype frequencies		
MM	rdl296c_rdl296c_n	Number of mosquitoes in the sample with the 296C/296C genotype (homozygous susceptible (SS)).
MN	rdl296c_rdl296c_percent	Frequency (percent) of mosquitoes in the sample with the 296C/296C genotype.
MO	rdl296c_rdl296g_n	Number of mosquitoes in the sample with the 296C/296G

		genotype (heterozygous resistant (RS)).
MP	rdl296c_rdl296g_percent	Frequency (percent) of mosquitoes in the sample with the 296C/296G genotype.
MQ	rdl296g_rdl296g_n	Number of mosquitoes in the sample with the 296G/296G genotype (homozygous resistant (RR)).
MR	rdl296g_rdl296g_percent	Frequency (percent) of mosquitoes in the sample with the 296G/296G genotype.
MS	rdl296c_rdl296s_n	Number of mosquitoes in the sample with the 296C/296S genotype (heterozygous resistant (RS)).
MT	rdl296c_rdl296s_percent	Frequency (percent) of mosquitoes in the sample with the 296C/296S genotype.
MU	rdl296s_rdl296s_n	Number of mosquitoes in the sample with the 296S/296S genotype (homozygous resistant (RR)).
MV	rdl296s_rdl296s_percent	Frequency (percent) of mosquitoes in the sample with the 296S/296S genotype.
MW	rdl296g_rdl296s_n	Number of mosquitoes in the sample with the 296G/296S genotype (homozygous resistant (RR)).
MX	rdl296g_rdl296s_percent	Frequency (percent) of mosquitoes in the sample with the 296G/296S genotype.
Rdl296 allele frequencies		
MY	rdl296c_percent	Frequency (percent) of mosquitoes in the sample with the 296C (susceptible) allele.
MZ	rdl296g_percent	Frequency (percent) of mosquitoes in the sample with the 296G (resistant) allele.
NA	rdl296s_percent	Frequency (percent) of mosquitoes in the sample with the 296S (resistant) allele.

Altered Acetylcholinesterase (ace-1) marker

This section records the acetylcholinesterase target site mutation: Ace1-G280S (formally named Ace1-G119S).

Table 23: Altered Acetylcholinesterase columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
NB	ace1_method_1	Method used for the altered acetylcholinesterase (ace-1) mutation test in the format Author Year of publication.
NC	ace1_number_of_mosquitoes_tested	Number of mosquitoes tested for ace-1 resistance.
ND	ace1_generation	The mosquito generation tested: F0, F1 or a mix of both.
Ace1-280 genotype frequencies		
NE	ace1_280g_ace1_280g_n	Number of mosquitoes in the sample with the 280/280 genotype (homozygous susceptible (SS)).
NF	ace1_280g_ace1_280g_percent	Frequency (percent) of mosquitoes in the sample with the 280/280 genotype.
NG	ace1_280g_ace1_280s_n	Number of mosquitoes in the sample with the 280/280S genotype (heterozygous resistant (RS)).
NH	ace1_280g_ace1_280s_percent	Frequency (percent) of mosquitoes in the sample with the 280/280S genotype.
NI	ace1_280s_ace1_280s_n	Number of mosquitoes in the sample with the 280/280S genotype (homozygous resistant (RR)).

NJ	ace1_280s_ace1_280s_percent	Frequency (percent) of mosquitoes in the sample with the 280/280S genotype.
<i>Ace1-280 allele frequencies</i>		
NK	ace1_280g_percent	Frequency (percent) of mosquitoes in the sample with the 280G (susceptible) allele.
NL	ace1_280s_percent	Frequency (percent) of mosquitoes in the sample with the 280S (resistant) allele.

Glutathione S-transferases (Gste) markers

This section records two resistance mutations of Glutathione S-transferases: Gste2-114 and Gste2-119.

Table 24: Glutathione S-transferases (GSTe) columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
NM	gste_method_1	Method used for the Glutathione S-transferases (Gste) mutation test in the format Author Year of publication.
NN	gste_number_of_mosquitoes_tested	Number of mosquitoes tested for GSTe resistance.
NO	gste_generation	The mosquito generation tested: F0, F1 or a mix of both.
<i>Gste2-114 genotype frequency</i>		
NP	gste2_114l_gste2_114l_n	Number of mosquitoes in the sample with the 114L/114L genotype (homozygous susceptible (SS)).
NQ	gste2_114l_gste2_114l_percent	Frequency (percent) of mosquitoes in the sample with the 114L/114L genotype.
NR	gste2_114l_gste2_114t_n	Number of mosquitoes in the sample with the 114L/114T genotype (heterozygous resistant (RS)).
NS	gste2_114l_gste2_114t_percent	Frequency (percent) of mosquitoes in the sample with the 114L/114T genotype.
NT	gste2_114t_gste2_114t_n	Number of mosquitoes in the sample with the 114T/114T genotype (homozygous resistant (RR)).
NU	gste2_114t_gste2_114t_percent	Frequency (percent) of mosquitoes in the sample with the 114T/114T genotype.
<i>Gste2-114 allele frequency</i>		
NV	gste2_114l_percent	Frequency (percent) of mosquitoes in the sample with the 114L (susceptible) allele.
NW	gste2_114t_percent	Frequency (percent) of mosquitoes in the sample with the 114T (resistant) allele.
<i>Gste2-119 genotype frequency</i>		
NX	gste2_119l_gste2_119l_n	Number of mosquitoes in the sample with the 119/119 genotype (homozygous susceptible (SS)).
NY	gste2_119l_gste2_119l_percent	Frequency (percent) of mosquitoes in the sample with the 119/119 genotype.
NZ	gste2_119l_gste2_119v_n	Number of mosquitoes in the sample with the 119/119V genotype (heterozygous resistant (RS)).
OA	gste2_119l_gste2_119v_percent	Frequency (percent) of mosquitoes in the sample with 119/119V genotype.
OB	gste2_119v_gste2_119v_n	Number of mosquitoes in the sample with the 119V/119V genotype (homozygous resistant (RR)).
OC	gste2_119v_gste2_119v_percent	Frequency (percent) of mosquitoes in the sample with

		119V/119V genotype.
	<i>Gste2-119 allele frequency</i>	
OD	gste2_119l_percent	Frequency (percent) of mosquitoes in the sample with the 119 (susceptible) allele.
OE	gste2_119v_percent	Frequency (percent) of mosquitoes in the sample with the 119V (resistant) allele.

Cytochrome P450 (Cyp) markers

This section records cytochrome p450s (Cyp or p450s) genetic markers: Cyp4j5, Cyp6p4 and Cyp6aap.

Table 25: Cytochrome p450s (Cyp or p450s) columns in the VADB, with descriptions.

	COLUMN HEADING	DESCRIPTION
OF	cyp_method_1	Method used for the Cytochrome p450s (Cyp) mutation test in the format Author Year of publication.
OG	cyp_number_of_mosquitoes_tested	Number of mosquitoes tested for Cyp resistance.
OH	cyp_generation	The mosquito generation tested: F0, F1 or a mix of both.
	<i>Cyp4j5-43 genotype frequency</i>	
OI	cyp4j5_43l_cyp4j5_43l_n	Number of mosquitoes in the sample with the Cyp4j5-L43L/ Cyp4j5-L43L genotype (homozygous susceptible (SS)).
OJ	cyp4j5_43l_cyp4j5_43l_percent	Frequency (percent) of mosquitoes in the sample with the Cyp4j5-L43L/ Cyp4j5-L43L genotype.
OK	cyp4j5_43l_cyp4j5_43f_n	Number of mosquitoes in the sample with the Cyp4j5-L43L/ Cyp4j5-L43F genotype (heterozygous resistant (RS)).
OL	cyp4j5_43l_cyp4j5_43f_percent	Frequency (percent) of mosquitoes in the sample with the Cyp4j5-L43L/ Cyp4j5-L43F genotype.
OM	cyp4j5_43f_cyp4j5_43f_n	Number of mosquitoes in the sample with the Cyp4j5-L43F/ Cyp4j5-L43LF genotype (homozygous resistant (RR)).
ON	cyp4j5_43f_cyp4j5_43f_percent	Frequency (percent) of mosquitoes in the sample with the Cyp4j5-L43F/ Cyp4j5-L43LF genotype.
	<i>Cyp4j5-43 allele frequency</i>	
OO	cyp4j5_43l_percent	Frequency (percent) of mosquitoes in the sample with the Cyp4j5-L43L (susceptible) allele.
OP	cyp4j5_43f_percent	Frequency (percent) of mosquitoes in the sample with the Cyp4j5-L43F (resistant) allele.
	<i>Cyp6p4-236 genotype frequency</i>	
OQ	cyp6p4_236wt_cyp6p4_236wt_n	Number of mosquitoes in the sample with the Cyp6p4-236WT/ Cyp6p4-236WT genotype (homozygous susceptible (SS)).
OR	cyp6p4_236wt_cyp6p4_236wt_percent	Frequency (percent) of mosquitoes in the sample with Cyp6p4-236WT/ Cyp6p4-236WT genotype.
OS	cyp6p4_236wt_cyp6p4_236m_n	Number of mosquitoes in the sample with the Cyp6p4-236WT/ Cyp6p4-236M genotype (heterozygous resistant (RS)).
OT	cyp6p4_236wt_cyp6p4_236m_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6p4-236WT/ Cyp6p4-236M genotype.
OU	cyp6p4_236m_cyp6p4_236m_n	Number of mosquitoes in the sample with the Cyp6p4-236M/ Cyp6p4-236M genotype (homozygous resistant (RR)).

OV	cyp6p4_236m_cyp6p4_236m_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6p4-236M/ Cyp6p4-236M genotype.
<i>Cyp6p4-236 allele frequency</i>		
OW	cyp6p4_236wt_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6p4-236WT (susceptible) allele.
OX	cyp6p4_236m_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6p4-236M (resistant) allele.
<i>Cyp6aap genotype frequency</i>		
OY	cyp6aap_wt_cyp6aap_wt_n	Number of mosquitoes in the sample with the Cyp6aap-WT/Cyp6aap-WT genotype (homozygous susceptible (SS)).
OZ	cyp6aap_wt_cyp6aap_wt_percent	Frequency (percent) of mosquitoes in the sample with the of Cyp6aap-WT/Cyp6aap-WT genotype.
PA	cyp6aap_wt_cyp6aap_dup1_n	Number of mosquitoes in the sample with the Cyp6aap-WT/Cyp6aap-Dup1 genotype (heterozygous resistant (RS)).
PB	cyp6aap_wt_cyp6aap_dup1_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6aap-WT/Cyp6aap-Dup1 genotype.
PC	cyp6aap_dup1_cyp6aap_dup1_n	Number of mosquitoes in the sample with the Cyp6aap-Dup1/Cyp6aap-Dup1 genotype (homozygous resistant (RR)).
PD	cyp6aap_dup1_cyp6aap_dup1_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6aap-Dup1/Cyp6aap-Dup1 genotype.
<i>Cyp6aap allele frequency</i>		
PE	cyp6aap_wt_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6aap-WT (susceptible) allele.
PF	cyp6aap_dup1_percent	Frequency (percent) of mosquitoes in the sample with the Cyp6aap-Dup1 (resistant) allele.

Contact us

If you have any feedback or would like further information, please do not hesitate to contact us at: vectoratlas@icipe.org.